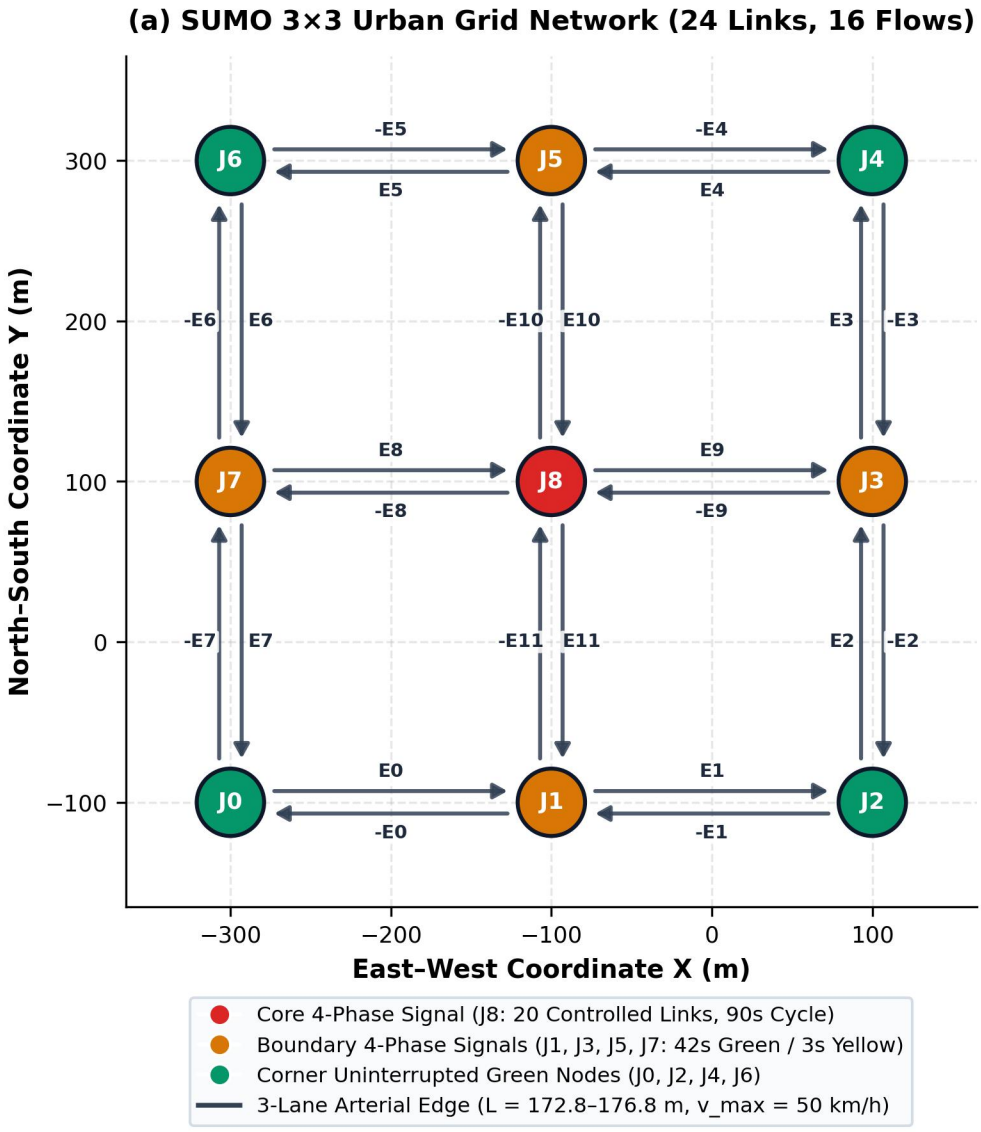


(b) Closed-Loop Physics-Informed RL-CACC Architecture

1. SUMO Mixed Traffic & V2X Environment

- 50% Human IDM (τ = 1.4 s, σ = 0.5, s_0 = 2.5 m)
- 50% Automated Fleet (ACC / CACC / RL / RL-CACC)
- V2V Sidelink: Leader $v_l(t - \tau_d)$, $a_l(t - \tau_d)$
- Onboard Radar/LiDAR: Gap g_t, Rel. Speed Δv_t
- 16 OD Flows (4,800 veh/h, 90 s Signal Cycles)

TraCI + V2V

2. Augmented State Vector s_t

- Ego Kinematics: Speed v_t, Accel a_t
- Spacing & Headway: g_t, $h_t = g_t / \max(v_t, 0.1)$
- Cooperative V2V States: Δv_t, $v_{l,t}$, $a_{l,t}$
- Signal SPaT Context: Dist. d_{tls}, Phase φ_{tls}
- Normalized Input Observation $s_t \in \mathbb{R}^5$–$\mathbb{R}^7$

State s_t

3. PPO Actor-Critic & Safety Shield

- Policy $\pi_\theta(a_t \mid s_t)$ & Value $V_\phi(s_t)$ (MLP 64×64)
- Clipped Objective $L^{CLIP}(\theta)$ + GAE (λ = 0.95)
- Speed Adjustment Action: $u_t \in [-1.0, 1.0]$ m/s
- Gipps / CBF Safety Filter & Actuator Lag:
  $v_{t+1} = \min(v_{safe}, \max(0, v_t + u_t))$

Experience

4. Multi-Objective Reward Function r_t

- Mobility Gain: $+0.7\, v_t$ (throughput efficiency)
- Stability Damping: $-0.5\, |\Delta v_t|$ (wave attenuation)
- Safety Barrier: $-1.0 \max(0, 5 - g_t)$ (headway)
- Smoothness Cost: $-0.3\, |a_t|$ (jerk suppression)
- Total: $r_t = 0.7 v_t - 0.5|\Delta v_t| - \max(0,5-g_t) - 0.3|a_t|$

PPO Update

v_{t+1}